

Road Trip BOS-FLO Project

OR 6205: Deterministic Operations Research

Individual Project Report Rubric

Purpose of Individual Reports

While you worked collaboratively as a team, each student will submit an individual report. This report should demonstrate YOUR understanding of the modeling choices, analysis, and recommendations your team developed together. You are not simply summarizing what the group did, but you are showing that you can explain and defend the work as if you had done it yourself. This clarity helps students understand expectations and enhances report quality.

Grading Overview

Criterion	Weight	Focus
Decision and Assumption Justification	25%	Can you explain WHY choices were made?
Mathematical Model Appropriateness	25%	Do you understand the model and its fit to the problem?
Results Interpretation and Sensitivity Analysis	20%	Can you explain what the results MEAN?
Recommendations	15%	Can you explain what the results MEAN?
Critical Analysis and Limitations	5%	Do you recognize consequences and blind spots?
Communication Effectiveness	10%	Is your report clear, professional, and complete?

Full Explanation of each Criterion

Criterion 1: Decision and Assumption Justification (25 points)

This criterion assesses whether you understand the reasoning behind your team's modeling choices, not just what was chosen, but why.

Excellent (23-25 points)

For each major modeling choice, you demonstrate the "X instead of Y and Z" reasoning: you explain what was chosen, what alternatives existed, why this choice fits the problem and decision-maker's needs, and what consequences follow from this choice. You clearly distinguish between assumptions you made by choice versus constraints imposed externally (time, data availability, scope). Your justifications show that you personally understand the trade-offs, not just that you can repeat what the team decided. A reader would trust you to make similar decisions independently on a future project.

Proficient (18-22 points)

You explain most modeling choices with reasoning tied to the problem context. Some justifications fully address alternatives and consequences, while others provide a partial explanation. You acknowledge the difference between chosen assumptions and external constraints, though this distinction may not be consistent throughout. Your understanding is evident, though occasional gaps suggest you may be more confident with some aspects of the model than others.

Developing (13-17 points)

You describe what choices were made, but provide a limited explanation of why. Alternatives are mentioned briefly, if at all. The consequences of choices are stated in general terms rather than linked to specific problem features or stakeholder impacts. The distinction between chosen assumptions and external constraints is unclear or missing. A reader might wonder whether you fully understand the reasoning or are reporting decisions made by teammates.

Beginning (0-12 points)

Choices are listed without meaningful justification. The report reads more like a description of what was done than an explanation of why. Little evidence that you understand the reasoning behind the team's decisions. Assumptions may be unstated or presented without acknowledgment that alternatives existed.

Criterion 2: Mathematical Model Appropriateness (25 points)

This criterion assesses whether you understand why your chosen modeling approach fits the problem, and whether you could recognize when a different approach would be needed.

Excellent (23-25 points)

You clearly explain how your model structure (objective function, decision variables, constraints) connects to the real-world problem. You articulate why this type of model (LP, network, etc.) is appropriate given the problem characteristics and available data. You discuss other modeling approaches your team considered and explain specifically why they were less suitable, not just "we chose LP" but "we chose LP rather than [alternative] because [specific reason related to problem structure]." Your explanation shows that you can select an appropriate model for a new problem, not just the two that the group created.

Proficient (18-22 points)

You explain the connection between your model and the problem, with clear descriptions of how decision variables, objective, and constraints represent real-world elements. You mention alternative approaches, though the reasoning for your choice over alternatives may be general rather than specific to problem characteristics. Your understanding of model appropriateness is solid, though you could improve at articulating why alternatives were rejected.

Developing (13-17 points)

You present the model formulation but provide a limited explanation of why this structure fits the problem. The connection between mathematical elements and real-world meaning is implicit rather than explicit. Alternative approaches are neither discussed nor dismissed with substantive reasoning. A reader might understand what model you used, but not why it was the right choice.

Beginning (0-12 points)

The model is presented with minimal or no explanation of its appropriateness. Mathematical formulation may contain errors or inconsistencies, and there is no discussion of alternatives. The reader cannot determine whether you understand why this modeling approach fits (or doesn't fit) the problem.

Criterion 3: Results Interpretation and Sensitivity Analysis (20 points)

This criterion assesses whether you can explain what the results mean and how confident decision-makers should be in those results.

Excellent (18-20 points)

You present results clearly and interpret them in context, not just "the optimal value is X" but "this means [specific interpretation relevant to the problem]." Your sensitivity analysis is purposeful: you explain why you tested the parameters you tested (connecting to uncertainty in the real world), what you learned from the analysis, and what this means for how much confidence to place in the results. You address the solution's robustness and the conditions under which the optimal solution would change. A reader understands both the results and their reliability.

Proficient (14-17 points)

You present results with an interpretation that connects to the problem context. Sensitivity analysis is included, with an explanation of which parameters were tested and the findings. The connection between parameter uncertainty and real-world conditions is present but may not be fully developed. A reader understands the results, though might have questions about specific aspects of robustness.

Developing (10-13 points)

Results are presented, but interpretation is limited or generic. Sensitivity analysis is included, but the choice of parameters to test is not well justified, and the findings are reported without explanation of their implications. The reader understands the numerical results but not fully what they mean.

Beginning (0-9 points)

Results are incomplete, incorrectly presented, or presented without interpretation. Sensitivity analysis is missing, superficial, or disconnected from the problem context.

Criterion 4: Recommendations (15 points)

This criterion assesses whether you can translate your analysis into actionable guidance for decision-makers.

Excellent (14-15 points)

Your recommendations flow logically from your results and sensitivity analysis. You clearly state the action the decision-maker should take and why, grounding your recommendation in the evidence from your analysis. You include appropriate caveats—conditions under which the recommendation applies, factors the decision-maker should monitor, or circumstances that would warrant revisiting the analysis. Your recommendations acknowledge the stakeholders affected and consider implementation realities. A decision-maker reading your report would know what to do, why to do it, and what to watch for.

Proficient (11-13 points)

Your recommendations connect to your results and are actionable. You provide reasoning for your recommendations, though the connection to specific findings may not always be explicit. Some caveats or conditions are mentioned, but the implementation guidance may be general. A decision-maker would understand the recommended action, though might need to infer some details about when or how to apply it.

Developing (7-10 points)

Recommendations are present but weakly connected to the analysis. The reader can identify what you suggest, but not clearly why the analysis supports this recommendation over alternatives. Caveats and conditions are minimal or absent. Stakeholder considerations are not addressed. A decision-maker might question whether this recommendation is well-supported.

Beginning (0-6 points)

Recommendations are missing, vague, or disconnected from the analysis. No evidence that you can translate modeling results into guidance for action. A decision-maker could not act on this report.

Criterion 5: Critical Analysis and Limitations (5 points)

This criterion assesses your ability to think beyond the model to recognize consequences, limitations, and blind spots.

Excellent (5 points)

You demonstrate genuine critical thinking about the model and its recommendations. You identify potential unintended consequences and consider who might be affected positively or negatively. You acknowledge limitations honestly: what the model cannot capture, data gaps, and assumptions that might not hold. You include a reflection on your team's potential blind spots and how your collective perspectives might have shaped the analysis. You identify what questions you would explore with more time or data.

Proficient (4 points)

You acknowledge limitations and discuss potential consequences of the recommendations. Blind-spot reflection is present and genuine, though it may be brief. You identify what the model cannot capture, though the discussion may be general rather than specific. Your analysis goes beyond description but may not fully explore the implications.

Developing (3 points)

Limitations are mentioned but treated superficially, listed rather than analyzed, and there is little discussion of unintended consequences or stakeholder impacts. Blind spots reflection is absent or perfunctory. The report describes what was done but does not critically examine what could go wrong.

Beginning (0-3 points)

No meaningful discussion of limitations, consequences, or blind spots. The report presents the model and results as if they are complete and unproblematic.

Criterion 6: Communication Effectiveness (10 points)

This criterion assesses whether your report is clear, professional, and accessible.

Excellent (9-10 points)

Your report is well organized with clear section headings that guide the reader. Technical terms are defined when first used. Visual elements (tables, figures, diagrams) are purposeful, numbered, titled, and referenced in the text. Language is professional. The report passes the "new team member" test: someone unfamiliar with your project could understand your approach and continue your work.

Proficient (7-8 points)

The report is organized and readable with appropriate sections. Most technical terms are defined. Visual elements are included and generally support understanding. Language is professional with minor lapses. A new reader could follow most of the report.

Developing (4-6 points)

Organization is present but inconsistent. Technical terms may be used without definition. Visual elements are present but not well integrated. A new reader would have significant questions about structure or meaning.

Beginning (0-3 points)

The report lacks clear organization. Technical terms are undefined. Visual elements are missing or confusing. A new reader would struggle to understand the work.

Incorporation of Feedback

Your report must include changes requested by the instructor during the final presentation. Reports that do not address this feedback will receive a deduction of up to 10 points, depending on the significance of the unaddressed feedback.

Required Elements Checklist

Before submitting, confirm your report includes:

- **Clear statement of the problem you're solving**
- **Final model assumptions (justified)**
- **Results with interpretation**
- **Sensitivity analysis with explanation of choices and findings**
- **Recommendations of action with appropriate caveats**
- **Limitations and blind spots reflection**
- **Changes addressing instructor feedback from the presentation**

A Note on Individual Understanding

Because this is an individual report based on collaborative work, I am looking for evidence that YOU understand the full project, not just the parts you personally worked on. Strong reports demonstrate that you can explain any aspect of the model, defend any assumption, and interpret any result.

This is not about catching anyone. It's about ensuring that collaborative work genuinely builds everyone's capabilities. The goal is for each of you to leave this project able to tackle similar problems independently.

Disclaimer: I used Claude AI to help me create the rubric, based on the baseline of what I wanted to grade and why it was important. I've edited it, and I am fully responsible for its content.

Questions about expectations? My door is always open. It is better to ask for clarification before submitting than to misunderstand the expectations.